



Pulumi

Introduction to Pulumi IaC with AWS in Python

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Community Engineer, Pulumi

Housekeeping

Be chatty in the **chat tab**

Ask questions in the **Q&A tab**

Links to slides, demos, etc. are in the **handouts tab**

This session is being recorded. Link to video in follow-up email.

Today's Agenda

Pulumi IaC Quick Intro

The Pulumi Platform and Common Questions

Demo: AWS Lambda

Q&A

Wrap Up

Prerequisites

Pulumi CLI installed

Python 3.X installed

Full Admin Access to an AWS account

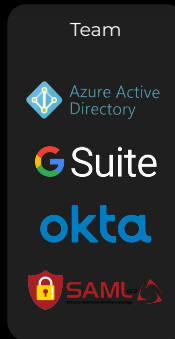
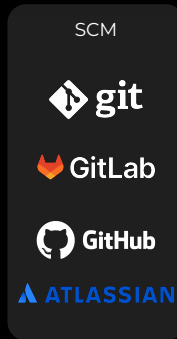
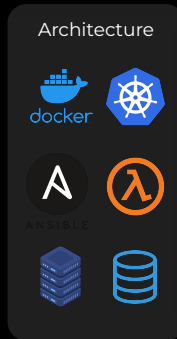
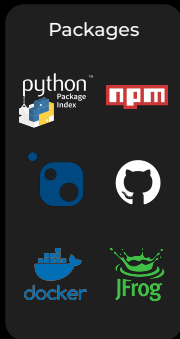
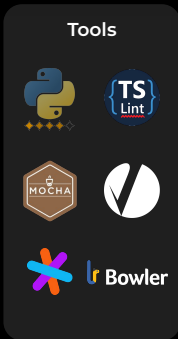
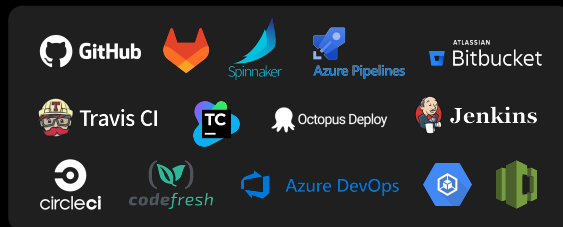
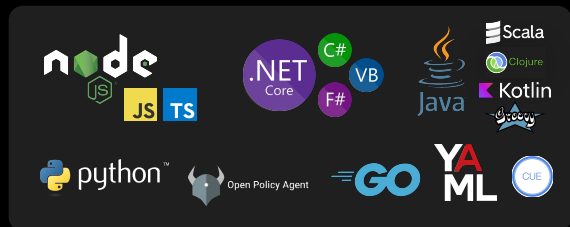
Sign up for a free Pulumi Cloud Account



Pulumi IaC in a nutshell

Pulumi IaC

Your cloud, your language, your way



170+ Clouds Supported

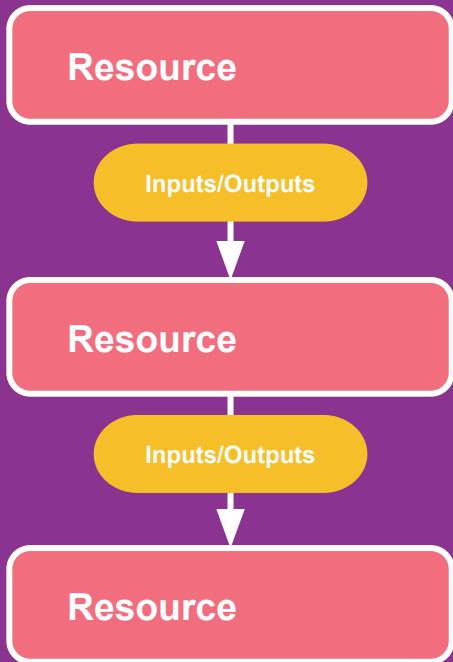
IaC Product Comparison



	Pulumi	Terraform	CDK	CloudFormation
Language(s)	General purpose (multiple + YAML)	HCL (or JSON)	General purpose (multiple)	YAML (or JSON)
Resources	Any cloud	Any cloud	AWS only	AWS only
Deployment Model	Local CLI	Local CLI	Synth to YAML, CloudFormation Service	CloudFormation Service
Secrets Handling	Encrypted in transit and rest, state file and config file (or Pulumi ESC)	Ephemeral values (not stored)	AWS Secrets Manager	AWS Secrets Manager

Project

Program



Stack (DEV)

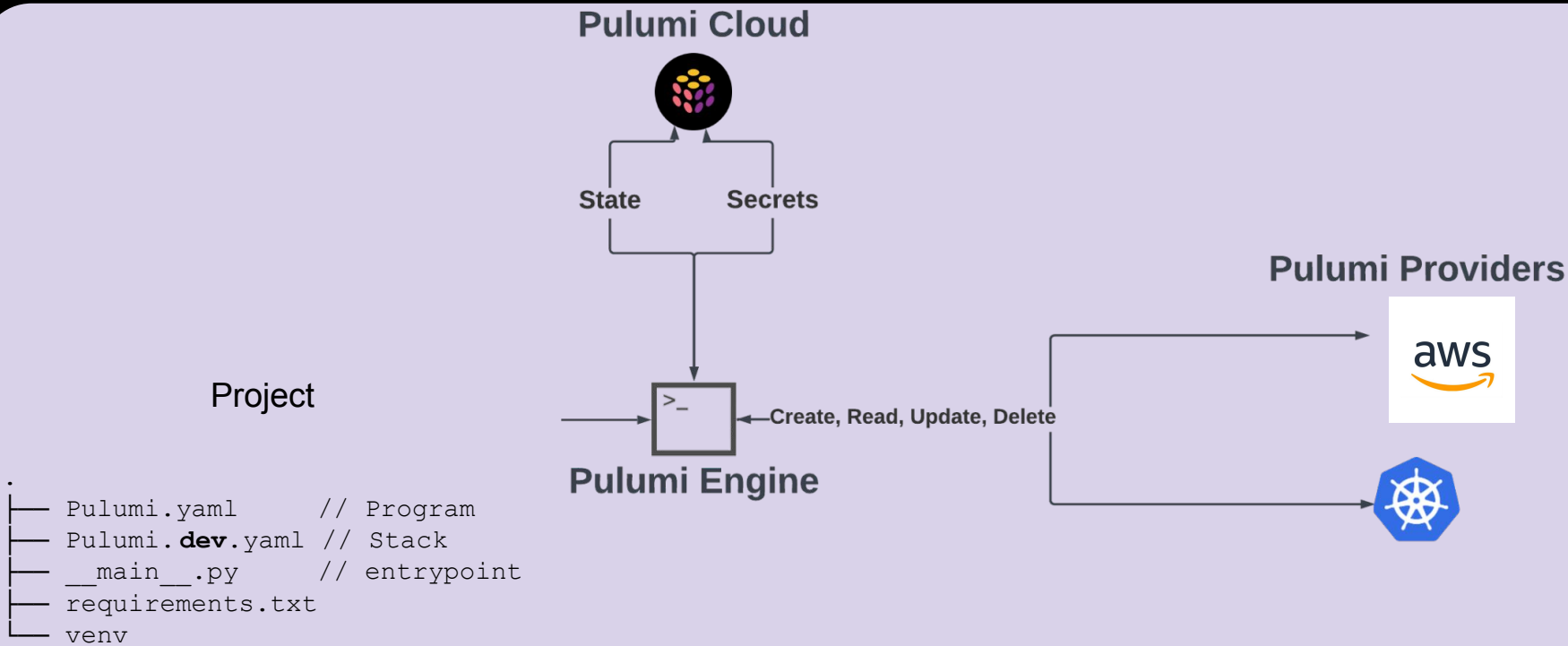
Stack (QA)

Stack (PROD)

Pulumi Program Model



Pulumi IaC Model





Demo: Authoring Pulumi Programs

Code

Code:

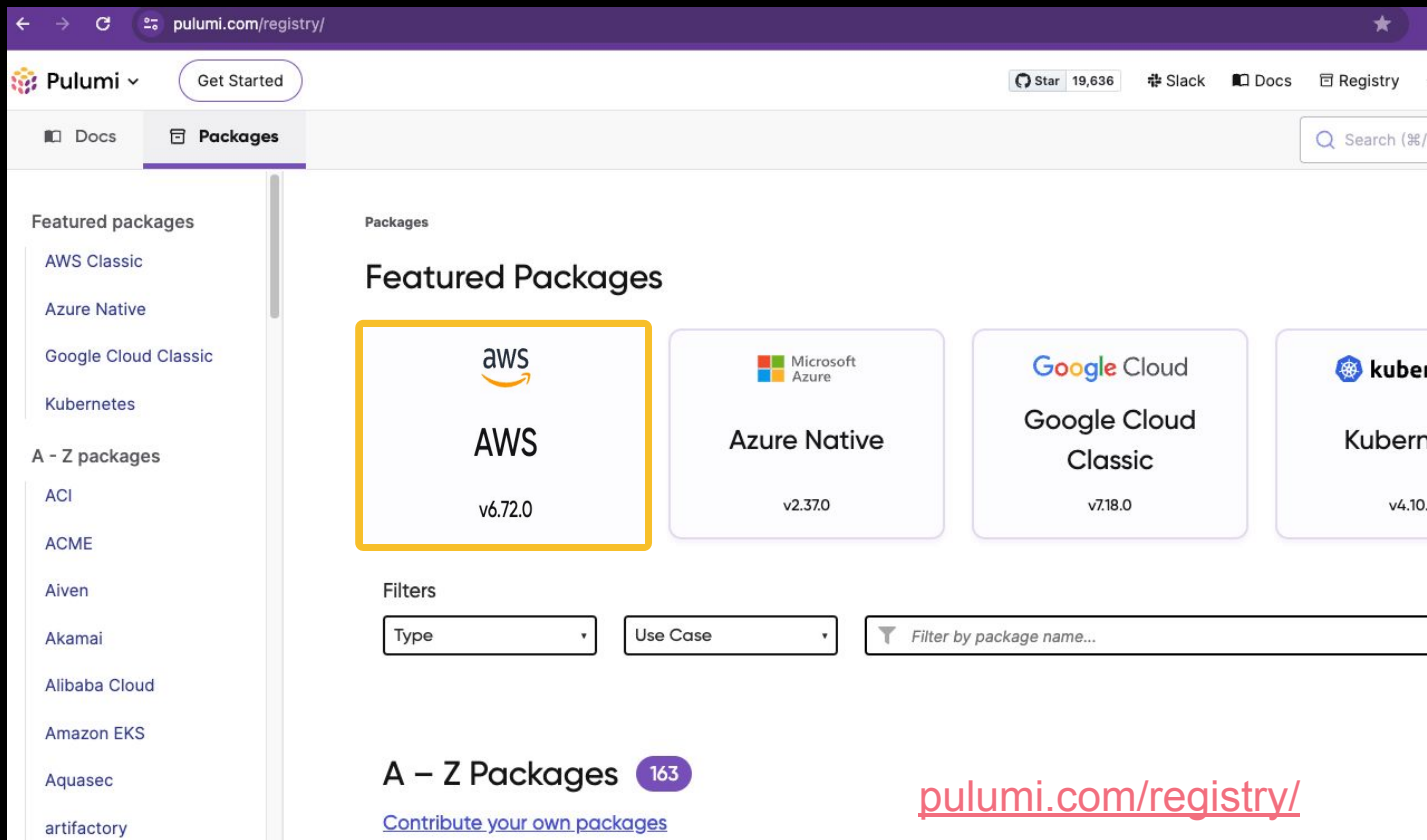
<https://github.com/pulumi/workshops/tree/main/aws-intro-python>

Blog Version: <https://www.pulumi.com/blog/serverless-api/>



The Pulumi Platform and Common Questions

Pulumi registry



The screenshot shows the Pulumi registry website at pulumi.com/registry/. The page features a navigation bar with the Pulumi logo, a "Get Started" button, and links for "Star" (19,636), "Slack", "Docs", and "Registry". Below the navigation bar, there are tabs for "Docs" and "Packages", with "Packages" being the active tab. A search bar is located on the right side of the "Packages" tab.

On the left side, there is a sidebar with "Featured packages" and "A - Z packages". The "Featured packages" list includes AWS Classic, Azure Native, Google Cloud Classic, and Kubernetes. The "A - Z packages" list includes ACI, ACME, Aiven, Akamai, Alibaba Cloud, Amazon EKS, Aquasec, and artifactory.

The main content area is titled "Featured Packages" and displays four package cards:

- AWS** (v6.72.0) - This card is highlighted with a yellow border. It features the AWS logo and the text "AWS v6.72.0".
- Azure Native** (v2.37.0) - Features the Microsoft Azure logo and the text "Azure Native v2.37.0".
- Google Cloud Classic** (v7.18.0) - Features the Google Cloud logo and the text "Google Cloud Classic v7.18.0".
- Kubernetes** (v4.10.0) - Features the Kubernetes logo and the text "Kubernetes v4.10.0".

Below the featured packages, there is a "Filters" section with three dropdown menus: "Type", "Use Case", and "Filter by package name...".

At the bottom of the page, there is a section titled "A - Z Packages" with a badge indicating 163 packages. Below this, there is a link to "Contribute your own packages".

The URL pulumi.com/registry/ is displayed in red text at the bottom right of the page.



Common Questions

How can I migrate to Pulumi?
What is Pulumi Cloud?

Fun Links

Pulumi Registry: <https://www.pulumi.com/registry/>

Pulumi Concepts: <https://www.pulumi.com/docs/iac/concepts/>



Questions?



Wrapping up

PulumiUP 2025

- EMEA & Americas | May 6
 - 9 am ET
 - 2 pm GMT
 - 4 pm PT
- Asia Pacific | May 7 @ 9 am AEST



1

Discover the best practices and lessons learned on platform engineering.

2

Learn how to build modern infrastructure platforms to provide developer self-service.

3

Hear platform implementation stories from customers and watch demos of new product launches.

Register at <https://www.pulumi.com/pulumi-up/>

Continue your Pulumi journey

1. Join the Pulumi Community Slack!

Head to
<https://slack.pulumi.com/>
and join #aws and
#python

2. Sign up for Pulumi Cloud account and try it out!

Sign up for a free
account, or start a free
trial to check out
Pulumi's full set of
features!

3. Join us for an upcoming workshop!

Check your email!

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Thank you!

Adam Gordon Bell
Community Engineer
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The Pulumi Platform



AUTOMATE



Pulumi
laC

Everything is automated

Program the cloud - build internal platforms with familiar languages

Ensure best practices and governance are built-in



SECURE



Pulumi
ESC

Security is the default

Eliminate secrets and configuration sprawl at scale

Integrate and consume anywhere, with centralized management



MANAGE



Pulumi
Insights

Total visibility and controls

X-ray your entire cloud estate so you know everything you have

Gain insights, plan and drive changes, improve your posture

Pulumi ESC

ENVIRONMENTS, SECRETS, AND CONFIGURATIONS

- **Centralized management solution** for cloud applications and infrastructure secrets, and configurations
- Fully packed with enterprise features
- A managed service as part of Pulumi Cloud

Environments are YAML files contain a collection of secrets and configurations

```
values:
  1password:
    secrets:
      fn::open::1password-secrets:
        login:
          serviceAccountToken:
            fn::secret:
              ciphertext: ZXNje...
        get:
          google_oauth_id:
            ref: "op:///dev/app/u"
          google_oauth_secret:
            ref: "op:///dev/app/c"
```



Pulumi Deployments

Enables infrastructure CI/CD management at scale

- A managed service of Pulumi Cloud (available in all tiers,  included)
 - Multiple deployment triggers
 - git push, UI, CLI, Review Stacks, and  scheduling/TTL options,
 - Advanced controls
 - Webhook notifications, Policy checks,  Drift-detection/remediation options
- pulumi.com/docs/pulumi-cloud/deployments/

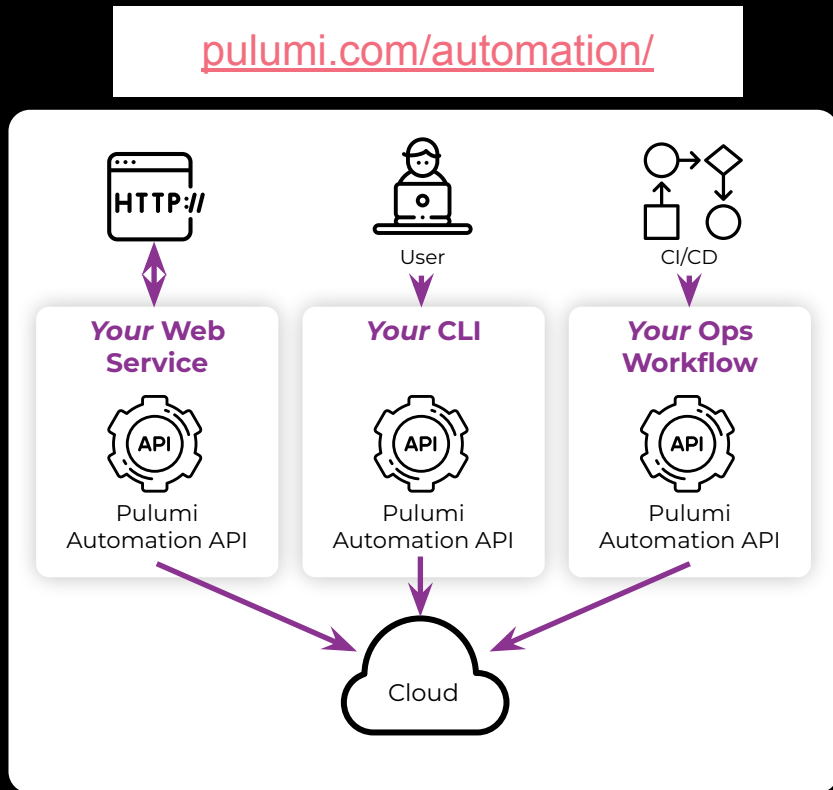


Automation API

- Use IaC as a library to
 - Orchestrate complex rollouts
 - Create self-service portals
 - Build custom tools and frameworks

Examples

- Build internal developer platforms
- Offer cloud service provisioning as part of their product





Pulumi AI

"I want a static website using Amazon S3 and CloudFront"

pulumi.com/ai

Here is a Pulumi program that defines these resources:

```
import * as pulumi from '@pulumi/pulumi';
import * as aws from '@pulumi/aws';

// Define the S3 bucket that will host the static website files.
const siteBucket = new aws.s3.Bucket('siteBucket', {
    // By specifying these website parameters, you configure the bucket for website hosting.
    website: {
        indexDocument: 'index.html', // The default page that S3 will serve
        errorDocument: '404.html',   // The error page that S3 will serve
    },
});

// Define a policy that allows public read of all objects in the bucket
const siteBucketPolicy = new aws.s3.BucketPolicy('siteBucketPolicy', {
    bucket: siteBucket.bucket, // reference the bucket created earlier
    policy: siteBucket.bucket.apply(publicReadPolicyForBucket), // apply the public read policy to the bucket
});

// Helper function that generates a public read policy for the S3 bucket
function publicReadPolicyForBucket(bucketName: string) {
```

TypeScript

JavaScript

Python

Go

C#

Java

YAML

Terraform

Thinking... ⚙️

GPT v4 v4-Turbo | Mode explain balanced code

🔗 Share conversation

